Multivariate Calculus Lecture Notes
Section 16.3 – Conservative Vector Fields

I. Review
   a. As we've said, a vector field $\mathbf{F}$ is *conservative* if there exists some scalar function V such that $\nabla V = \mathbf{F}$.
   b. If such a V exists it is called a *potential function* for $\mathbf{F}$.

II. Fundamental theorem of conservative vector fields
   a. Suppose $\mathbf{F}$ is a conservative vector field in D with potential function V. Suppose that C is a path from P to Q in D and that $\mathbf{r}(t)$ parametrizes the curve C such that $\mathbf{r}(a) = P$ and $\mathbf{r}(b) = Q$.
      i. Then we have

$$\int_C \mathbf{F} \cdot d\mathbf{r} = \int_a^b \nabla V(\mathbf{r}(t)) \cdot \mathbf{r}'(t) dt = \int_a^b \frac{dV(\mathbf{r}(t))}{dt}\, dt = V(\mathbf{r}(t))\Big|_a^b$$
$$= V(\mathbf{r}(b)) - V(\mathbf{r}(a)) = V(Q) - V(P).$$

      ii. Path independence: $\displaystyle\int_C \mathbf{F} \cdot d\mathbf{r} = V(Q) - V(P)$.
         1. Path independence says that if you want to evaluate a line integral in a conservative vector field you need only find a potential function, evaluate it at the starting and ending points of the curve, and subtract.
         2. This should remind you of the fundamental theorem of calculus; the potential function is playing the role of an antiderivative.
         3. In fact, path independence in $\mathbf{F}$ is equivalent to $\mathbf{F}$ having a potential function: $\mathbf{F}$ is conservative on a closed and connected domain D if and only if $\mathbf{F}$ is path independent on D.
      iii. Zero circulation: $\displaystyle\oint_C \mathbf{F} \cdot d\mathbf{r} = 0$.
         1. If the curve C is closed, then P and Q are the same point.
         2. Thus $V(P) = V(Q)$, and so we get $\displaystyle\int_C \mathbf{F} \cdot d\mathbf{r} = 0$.
   b. This is also sometimes called the *fundamental theorem of line integrals*.

III. Physical examples of conservative vector fields
   a. Gravitational fields and electric fields are conservative.
   b. Magnetic fields are not conservative.

IV. Show examples of vectors fields; ask whether they are conservative.

V. Examples
   a. Previous problem: Find the work done by the force $\mathbf{F}(x, y) = x^2\mathbf{i} - y\mathbf{j}$ over the curve $x = y^2$ from the point $(4, 2)$ to the point $(1, -1)$. (Problem 14.2 #19, Hass, Weir, and Thomas)
      i. After some work, we found the answer to be -19.5.
      ii. However, $\mathbf{F}$ is conservative, as $\dfrac{\partial F_1}{\partial y} = \dfrac{\partial F_2}{\partial x} = 0$.
      iii. Antidifferentiating F_1 and F_2 appropriately, we find

$$V(x, y) = \frac{1}{3}x^3 + g(y) = -\frac{1}{2}y^2 + h(x).$$

iv. Thus a potential function that works has $g(y) = -y^2/2$ and $h(x) = x^3/3$, so that $V(x, y) = x^3/3 - y^2/2$.

v. Therefore,

$$\int_C \mathbf{F} \cdot d\mathbf{r} = V(1, -1) - V(4, 2) = \frac{1}{3} - \frac{1}{2} - \left(\frac{64}{3} - \frac{4}{2}\right) = -\frac{63}{3} + \frac{3}{2} = -19.5.$$

b. Find $\displaystyle\int_C \mathbf{F} \cdot d\mathbf{r}$, where $\mathbf{F}(x, y) = (2x \sin y + e^x)\mathbf{i} + x^2 \cos y\mathbf{j}$ and C is the curve from $(1, 0)$ to $(0, \pi/2)$ along $y = \arccos x$.

i. Let's hope that $\mathbf{F}$ is conservative; otherwise, this is going to be an ugly integral.

ii. Test for conservative vector field

- We have $\dfrac{\partial F_1}{\partial y} = 2x \cos y$ and $\dfrac{\partial F_2}{\partial x} = 2x \cos y$.
- Thus we have a conservative vector field.

iii. A potential function is $V(x, y) = x^2 \sin y + e^x$.

iv. Thus $\displaystyle\int_C \mathbf{F} \cdot d\mathbf{r} = V(0, \pi/2) - V(1, 0) = 1 - e$.

c. Find $\displaystyle\int_C \mathbf{F} \cdot d\mathbf{r}$, where $\mathbf{F}(x, y, z) = (3z + 2xy)\mathbf{i} + (x^2 - 3y^2)\mathbf{j} + (3x + 2z)\mathbf{k}$ and C is the curve given by $\mathbf{r}(t) = e^t \sin t\mathbf{i} + e^t \cos t\mathbf{j} + t\mathbf{k}$, $0 \le t \le \pi$.

i. Test for conservative

1. We have $\dfrac{\partial F_1}{\partial z} = 3 = \dfrac{\partial F_3}{\partial x}$, $\dfrac{\partial F_1}{\partial y} = 2x = \dfrac{\partial F_2}{\partial x}$, $\dfrac{\partial F_2}{\partial z} = 0 = \dfrac{\partial F_3}{\partial y}$.
2. Thus $\mathbf{F}$ is conservative.

ii. A potential function is $V(x, y, z) = 3xz + x^2y - y^3 + z^2$.

iii. Thus $\displaystyle\int_C \mathbf{F} \cdot d\mathbf{r} = V(0, -e^\pi, \pi) - V(0, 1, 0) = e^{3\pi} + \pi^2 + 1$.

d. Find $\displaystyle\int_C \mathbf{F} \cdot d\mathbf{r}$, where $\mathbf{F}(x, y) = (x^3 + 1)\mathbf{i} + \sin(y^2)\mathbf{j}$, and C is the semicircular path from $(0, 0)$ to $(2, 0)$ along the top of a circle of radius 1 centered at $(1, 0)$.

i. For this problem we're going to step through three different options, two of which we have discussed thus far.

ii. Method 1 (the direct way): Find a parametrization of C and evaluate the line integral using the method of the previous section.

1. We have $\mathbf{r}(t) = (1 - \cos t)\mathbf{i} + \sin t\mathbf{j}$, $0 \le t \le \pi$.
2. Thus $\mathbf{r}'(t) = \sin t\mathbf{i} + \cos t\mathbf{j}$.
3. We have $\mathbf{F}(\mathbf{r}(t)) = ((1 - \cos t)^3 + 1)\mathbf{i} + \sin(\sin^2 t)\mathbf{j}$.
4. So the line integral is given by

$$\int_C \mathbf{F} \cdot d\mathbf{r} = \int_0^\pi \left(((1 - \cos t)^3 + 1) \sin t + \sin(\sin^2 t) \cos t\right)\, dt.$$

5. This is doable, but it will take some work (including breaking the integral into two integrals and using two different u-substitutions).

iii. Method 2: If the vector field is conservative, use the fundamental theorem of line integrals.

1. Conservative test: $\dfrac{\partial}{\partial y}(x^3 + 1) = \dfrac{\partial}{\partial x}(\sin(y^2))$? Yes, so $\mathbf{F}$ is conservative.
2. Unfortunately, we cannot antidifferentiate $\sin(y^2)$ with respect to y, so we cannot find a potential function.

3. Thus this method won't help us.

iv. Method 3, which we have not yet seen: If $\mathbf{F}$ is conservative, any path will work. So pick a better path than the one you're given.
   1. For the problem at hand, the direct path from $(0,0)$ to $(2,0)$ is simplest.
   2. This path is parametrized by $\mathbf{r}(t) = t\mathbf{i}$, $0 \le t \le 2$.
   3. We have $\mathbf{r}'(t) = \mathbf{i}$.
   4. Also, $\mathbf{F}(\mathbf{r}(t)) = (t^3+1)\mathbf{i}$.
   5. Thus $\displaystyle\int_C \mathbf{F}\cdot d\mathbf{r} = \int_0^2 \mathbf{F}(\mathbf{r}(t))\cdot\mathbf{r}'(t)\,dt = \int_0^2 (t^3+1)\,dt = \frac{1}{4}t^4 + t\Big|_0^2 =$ $4 + 2 - (0+0) = 6$.

v. Method 3 is the easiest for this problem.

Homework: Section 16.3 #2, 5, 8, 10, 12, 16, 17, 18, 20, 21.